

Classifier LSKS ... CS / S**Table of Contents****Designation****BA - Number**

Safety

BA 911093-00-4

Structure and Function

BA 911401-00-4

Maintenance

BA 911316-00-4

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1. General

These operating instructions Classifier - "Safety" shall be considered as a supplement to the Roller Grinding Mill - "Safety" instructions. It must be ensured that these operating instructions will be studied in detail prior to operation of the classifier.

2. Notes on Safety at Work

Particular attention shall be paid to the following notes as to safety at work:

- The inspection covers of the classifier and the classifier drive system may only be opened when all rotating parts of the classifier are standing still and the mill is not in operation.
- Protective measures have to be taken in order to avoid unintentional starting of the classifier or the mill.
- Protective devices for rotating parts must be properly installed before starting the machine.
- Adequate ventilation is required for maintenance work inside a classifier.

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1. Structure of the Classifier

The classifier can be divided into the following assembly groups

- classifier housing cpl.
- rotor cpl.
- classifier lining cpl.
- classifier drive cpl.
- flap positioning cpl.

The structure of these assembly groups is described in detail in the following chapters. The assembly groups are usually mounted at the location where the machine is erected. The type of joint used (screwed or welded) and the required tools are listed on the assembly drawings of the assembly groups.

2. Function of the Classifier

The LOESCHE studded basket classifier is an air-flow classifier. The ground and dried material flows into the classifier from below together with the carrier air.

The guide flaps integrated into the classifier lower housing redirect the dust/air mixture from an upward flow into a tangential flow. The ribbed rotor rotates in the same direction as the tangential flow.

In the gap between the guide flaps and the ribs of the rotor a field of centrifugal forces forms in which the dust/air mixture is classified. Larger (oversize) particles are flung outwards and sink under the influence of gravity. After leaving the gap, oversize particles are collected in a grit collector and are returned into the mill for repeated grinding. Finished material passes through the ribbed rotor together with the carrier air and leaves the classifier via the outlet pipe in the classifier upper housing.

The fineness of the finished material depends on the adjustment of the classifier flaps in the classifier lower housing and the rotor speed. Changing the adjustment of the flaps outwards (more tangential) and/or increasing rotor speed results, after classifying, in a finer finished material, lowering the rotor speed and moving the flaps inward (more radial) in a rougher material.

The flaps are held in pivots and can be adjusted by hand.

The direction in which the rotor turns depends on the flow vortex of the flaps. The directions of rotor rotation and the vortex of the blades are identical.

The conical coarse hopper creates an orderly circulation of coarse particles within the mill. On the one hand it separates the rising dust/air mixture from the coarse particles flowing back on its inside, and on the other hand it passes them into the middle of the bowl where they are reground.

3. Structure and Function of the Classifier Components

Classifier Housing cpl.

The assembly group "Classifier Housing cpl." mainly encompasses the static parts of the classifier. These are the classifier lower housing, the classifier upper housing with the outlet duct, the frame between the lower and upper part, the flaps and the grit funnel.

The feed chute for the mill is integrated into the sidewall of the lower housing. The cross section of the chute is externally round and continues inside the classifier as a semicircular trough.

In the case of a CS-classifier (for combined grinding of clinker and slag) the part of the chute in contact with the feed material is manufactured in hard-faced combination plate and welded into the lower housing. The chute is equipped with a cover inside the classifier for inspection and servicing work.

With S-classifiers (for grinding only slag) the chute is only manufactured in mild steel but equipped with a system of jets, connecting pipes and flanges which can be attached to a shock-blower unit. The air pulse generated by this frees the chute from possible caked-on feed material. This chute is equipped with two inspection covers, one of which is situated in the grit return cone.

The upper and lower classifier housing are welded to each other.

The frame is supported by the upper flange of the classifier lower housing. The frame serves to hold the guide flap axles and for attaching the multi-part flanged rings, which are for closing off the rotor and for stopping stray coarse particles. The mounting lugs for the clamping tubes of the rotor bearing cartridge are also attached to the inside to the frame. On the underside of the frame a deflector ring is situated between the dynamic rotor and the static flaps. This also helps to avoid the possibility of oversized particles bypassing the rotor.

The classifier upper housing is supported by the upper flange of the frame. This serves on the one hand to redirect and guide the dust/gas mixture flow (which leaves the rotor in a vertical direction) and on the other hand it supports the bearing for the classifier rotor and the driving unit.

Tubes with guide plates in the outlet duct serve both for reducing the vortex in the gas flow created within the rotor zone and also to support the classifier upper housing cover. An inspection cover in the classifier upper housing allows inspection and to gain access to this area.

Seals between the detachable flange connections prevent atmospheric air from flowing into the classifier, which is kept internally at underpressure during operation.

The grit funnel is located below the rotor. This is attached to the classifier lower housing by tubular supports. In the upper part of the grit funnel, angled brackets are bolted to ring segments, in which the lower axles of the guide vanes are held.

Flat steel components within the grit funnel below the rotor serve to reduce the rotary motion of the coarse particles and thereby the wear.

R o t o r c p l .

The assembly group "Rotor cpl." essentially consists of the rotor, the classifier blades, the sub-assembly group "Bearing Cartridge cpl.", supports and the lifting tackle to hang the rotor for maintenance.

The bearing cartridge passes centrally through the classifier upper housing. The attachment flange is welded to the classifier upper housing using a curved steel flat after the rotor has been adjusted. The clamping tubes serve to support the bearing unit in the area of the lower bearing. After rotor adjustment these are secured against twisting by spot welding in the area of the threaded stems. Additional sections of tube serve to protect them from wear.

The adjustment is made with the rotor attached to the bearing cartridge. The whole unit is aligned into the centre of the frame of the classifier casing with the aid of linear measurements and a precision water/spirit levelling device.

The attachment screws in the flange connections between rotor and intermediate shaft and between intermediate shaft and flanged bush of the bearing unit have to be tightened to the torque given on the drawing.

The classifier blades are pushed into the slots of the holder rings of the rotor and are secured against dropping out using profiled clamping pieces. To avoid impermissible imbalance the classifier blades have to be weighed individually. Classifier blades of equal weight are installed opposite to each other.

After the installation has been completed the rotor should be dynamically balanced to the requirements given in the drawing "Rotor cpl.".

The direction which the rotor turns is predetermined by the flow vortex created by the louvre ring and the positioning of the flaps. When the drive is switched off the rotor must be allowed to continue running freely until it has come to a full stop. A brake on the motor is not permissible.

According to the maintenance regulations all attachment screws must be regularly checked for tightness. Any ground material sticking to rotating parts must be removed immediately.

The rotor shaft is held within the bearing unit in roller bearings. The upper self-aligning thrust bearing bears the axial and radial forces, the lower self-aligning bearing only radial forces. The bearings are greased. The grease is supplied via two separate tubes.

The grease is supplied from below to the self-aligning thrust bearing, corresponding to the natural transport direction of its rollers. The grease leaves via the upper rotary shaft seal and forms there a grease collar which prevents dust from entering into the bearing chamber. It must be checked during inspections that a sufficiently large grease collar has formed. A disc of brass closes the bearing chamber at the bottom and prevents to a high degree the grease from being pressed into the free space in the bearing cartridge.

The grease for the lower bearing is supplied via a tube which is located within the bearing cartridge. Here too a disc of brass is arranged above the bearing to prevent the grease from entering into the free space in the bearing cartridge. At the bottom the bearing is sealed from the dusty classifying chamber by a rotary shaft seal and a labyrinth.

When they are exchanged it must be ensured that the rotary shaft seals are mounted the correct way up following the information given on the drawing.

Classifier Lining cpl.

The inside of the classifier lower housing is clad with plug-welded chequer plates as wear protection. The inner surface of the chute of the CS-classifiers is already armoured and thus requires no further wear protection. The outer surface of the chute is protected by a cladding of wear resistant material in the zone where it is exposed to the gas stream.

In the case of the S-classifier the chute is clad internally and externally with hard-faced compound plate. On the inner surface the cladding is slightly flatter than the true profile of the chute so that the air pulse initiated by the shock blowers can exit from the jets which lie beneath the cladding.

The supports of the grit return cone, the bearing cartridge and the connection piping of the shock blowers are all protected with angle irons formed out of hard-faced compound plate.

Lining of the Classifier Upper Housing cpl.

The classifier upper housing, including the outlet and parts of the bearing unit, are protected from wear with a special cement (see parts list).

Flap Positioning cpl.

This assembly group consists of the flap carrier cpl., the scale and the attachment and sealing material.

The flap plates of the assembly group "Classifier Housing cpl." are screwed to the flap carrier. When this is done care has to be taken that they are correctly arranged (see information on drawing).

The adjustment of the flaps and the scale which is used to display the angle of the flap is adjusted to is done following the LOESCHE working instructions "Mounting: Flap Adjustment at Classifier" (see parts list DK "Docu: Flap Positioning"). Necessary tools, in particular templates, are included in the assembly group "Auxiliaries Flap Positioning cpl.".

In these instructions and the necessary steps for adjusting the flaps are also described.

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Classifier Drive cpl.

This assembly group mainly comprises the components: motor, gearbox, couplings, coupling protection, supports, toothed disc and proximity switch.

The gearbox is supported by a support which is centred on and attached to the bearing cartridge of the assembly group "Rotor cpl." After adjusting the gearbox in relation to the rotor shaft, the gearbox is pinned in position following the information on the drawing. An flexible coupling transfers the torque from the gearbox to the rotor shaft.

The electric motor is adjusted during installation together with its support in relation to the shaft entering the gearbox. The support is then welded to the upper part of the classifier following the instructions on the drawing.

Another flexible coupling, enclosed by a coupling protection, connects the motor and gearbox.

The rotation of the classifier is monitored using a toothed disc pushed onto the rotor shaft and a proximity switch which has to be adjusted on site.

The rotor speed is adjusted according to the fineness of the material to be classified. It is done continuously by adjusting the motor frequency using a frequency converter and is possible during classifier operation.

The adjustment of the proximity switch is only allowed when the drive is stationary: **danger of injury** by rotating parts!



During commissioning and re-starting one has to make sure that the drive moves in the correct direction. An incorrect direction of rotation may result in overloading of the motor and in a reduced mill throughput.

Classifier Lubrication Line cpl.

This assembly group mainly encompasses the grease pump with storage container and two adjustable pump elements, pipings and couplings.

This unit serves to lubricate both ball bearings of the assembly group "Bearing Cartridge cpl.".

The grease pump is attached to the classifier upper housing or to the classifier drive platform using a mounting plate. The pump elements are connected to the grease piping shown in the drawings. The pump output can be regulated by rotating the adjustment collar and must be set to the maximum value.

The pump running time and required grease quantity can be taken from the "Lubrication Scheme: Classifier Lubrication".

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1. General Maintenance Information

Before commencing maintenance work the working safety information is to be observed (see chapter "Safety").

Carry out regular visual checks of the machine in order to spot any possible damage early.

Check all screw connections regularly for tightness. Treat screw threads with corrosion protection in order to avoid friction damage and corrosion.

Regularly check seals for problem free operation, renew if required.

2. Maintenance Information for Components**Rotor cpl., Flap Positioning cpl. and Classifier Drive cpl.**

The rotor blades and guide flaps are to be regularly inspected for wear. To this end, the inspection covers in the upper and lower housings must be opened, respectively removed.

In bigger classifiers with a rotor which can be entered, the replacement of worn blades and guide flaps must be done from the inside of the rotor. For replacing the guide flaps the rotor blades must be partially removed.

In smaller classifiers, where the rotor cannot be entered, to replace worn blades and guide flaps the rotor must be either moved down into the inside of the mill, or be lifted together with the classifier upper housing and the attached guide flap crown away from the classifier lower housing. For lowering the rotor the grit funnel which consists of several parts must be partially dismantled and the rollers must possibly be swung out.

When exchanging the blades the rotor should be sufficiently balanced. To enable this, the blades are individually weighed and sorted in such a way that every 2 blades with approximately the same weight are installed opposite to one another.

After changing the guide flaps and tightening the nuts, tack weld the nuts to prevent loosening.

Wear on the guide flap carrier can be repaired with a hard facing welding electrode.

If the flap carrier has to be exchanged, it must be realigned as described in works procedure "Installation and alignment of flap adjustment system".

For replacement of worn power belts in an LSKS...ZD and for initial and subsequent tensioning thereof see working instructions "Power belts pretension". The number may be found in the parts list "Classifier Drive".